

Numerical Methods

Week 14

Ordinary Differential Equations (ODEs)



ODE Outline

- Introduction to ODEs
- Taylor series methods
- Midpoint method
- **Runge-Kutta methods**

Runge-Kutta Methods



Learning Objectives

- ❑ To understand the motivation for using Runge-Kutta (RK) method and the basic idea used in deriving them
- ❑ To get familiar with Taylor series for functions of two variables
- ❑ To use RK method of order 2 and higher to solve ODEs.

Motivation

- ❑ We seek accurate methods to solve ODEs that do not require calculating high order derivatives
- ❑ The approach is to use a formula involving unknown coefficients then determine these coefficients to match as many terms of the Taylor series expansion as possible.

Runge-Kutta Method

Second Order Runge-Kutta (RK2)

$$K_1 = h f(t, x)$$

$$K_2 = h f(t + \alpha h, x + \beta K_1)$$

$$x(t + h) = x(t) + w_1 K_1 + w_2 K_2$$

Problem:

Find α, β, w_1, w_2

such that $x(t + h)$ is as accurate as possible.

Taylor Series in Two Variables

The Taylor Series discussed in Chapter 4 is extended to the 2-independent variable case.

This is used to prove RK formula.

Taylor Series in One Variable

The Taylor Series expansion of $f(x)$

$$f(x+h) = \sum_{i=0}^{n-1} \frac{h^i}{i!} f^{(i)}(x) + \frac{h^n}{n!} f^{(n)}(\bar{x})$$

Approximation

Error

where $\bar{x}$ is between x and $x+h$

Taylor Series in One Variable

- Another Look -

Define

$$\left(h \frac{d}{dx}\right)^i f(x) = h^i \frac{d^i f(x)}{dx^i} = f^{(i)}(x) h^i$$

The Taylor Series expansion of $f(x)$

$$f(x+h) = \sum_{i=0}^{n-1} \frac{1}{i!} \left(h \frac{d}{dx}\right)^i f(x) + \frac{1}{n!} \left(h \frac{d}{dx}\right)^n f(\bar{x})$$

$\bar{x}$ is between x and $x+h$

Definitions

Define

$$\left(h \frac{\partial}{\partial x} \right)^i f(x, y) = h^i \frac{\partial^i f}{\partial x^i}$$

$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^0 f(x, y) = f(x, y)$$

$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^1 f(x, y) = h \frac{\partial f(x, y)}{\partial x} + k \frac{\partial f(x, y)}{\partial y}$$

$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f(x, y) = h^2 \frac{\partial^2 f(x, y)}{\partial x^2} + 2hk \frac{\partial^2 f(x, y)}{\partial x \partial y} + k^2 \frac{\partial^2 f(x, y)}{\partial y^2}$$

Taylor Series Expansion

$$f(x, y) = (x + 1)(x + y + 2)^2$$

Partial derivatives evaluated at (0,0)

$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^0 f(x, y) \Big|_{(0,0)} = 4$$

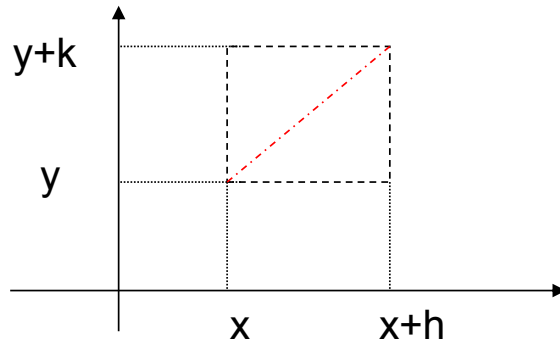
$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^1 f(x, y) \Big|_{(0,0)} = h f'_x + k f'_y \Big|_{(0,0)}$$

Taylor Series in Two Variables

The Taylor Series expansion of $f(x, y)$

$$f(x+h, y+k) = \sum_{i=0}^{n-1} \frac{1}{i!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^i f(x, y) + \underbrace{\frac{1}{n!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^n f(\bar{x}, \bar{y})}_{\text{error}}$$

$(\bar{x}, \bar{y})$ is on the line joining between (x, y) and $(x+h, y+k)$



Runge-Kutta Method

RK2

$$K_1 = h f(t, x)$$

$$K_2 = h f(t + \alpha h, x + \beta K_1)$$

$$x(t + h) = x(t) + w_1 K_1 + w_2 K_2$$

Problem:

Find α, β, w_1, w_2

such that $x(t + h)$ is as accurate as possible.

Runge-Kutta Method

Problem: Find α, β, w_1, w_2
to match as many terms as possible.

$$x(t+h) = x(t) + hx'(t) + \frac{h^2}{2} x''(t) + \frac{h^3}{6} x'''(t) + \dots$$

$$x(t+h) = x(t) + w_1 h f(t, x) + w_2 h f(t + \alpha h, x + \beta h f(t, x))$$

$$f(t + \alpha h, x + \beta h f) = f + \alpha h f'_t + \beta h f f'_x + \frac{1}{2} \left(\alpha h \frac{\partial}{\partial t} + \beta h \frac{\partial}{\partial x} \right)^2 f(\bar{t}, \bar{x})$$

$$x(t+h) = x(t) + (w_1 + w_2) h f(t, x) + \alpha w_2 h^2 f'_t + \beta w_2 h^2 f f'_x + O(h^3)$$

Runge-Kutta Method

$$x(t+h) = \boxed{x(t)} + \boxed{hx'(t)} + \boxed{\frac{h^2}{2}x''(t)} + \frac{h^3}{6}x'''(t) + \dots$$

$$x(t+h) = \boxed{x(t)} + \boxed{(w_1 + w_2)h f(t, x)} + \boxed{\alpha w_2 h^2 f'_t + \beta w_2 h^2 f f'_x} + O(h^3)$$

$$\Rightarrow w_1 + w_2 = 1, \quad \alpha w_2 = 0.5, \quad \beta w_2 = 0.5$$

One possible solution

$$w_1 = 0.5, \quad w_2 = 0.5, \quad \alpha = 1, \quad \beta = 1$$

Runge-Kutta Method

RK2

RK2

$$K_1 = h f(t, x)$$

$$K_2 = h f(t + h, x + K_1)$$

$$x(t + h) = x(t) + \frac{1}{2} (K_1 + K_2)$$

Runge-Kutta Method

Alternative Formula

RK2

$$F_1 = f(t, x)$$

$$F_2 = f(t + h, x + hF_1)$$

$$x(t + h) = x(t) + \frac{h}{2}(F_1 + F_2)$$

Runge-Kutta Method

Alternative Formula

RK2

$$K_1 = h f(t, x)$$

$$K_2 = h f(t + h, x + K_1)$$

$$x(t + h) = x(t) + \frac{1}{2}(K_1 + K_2)$$

Alternative Form

$$F_1 = f(t, x)$$

$$F_2 = f(t + h, x + hF_1)$$

$$x(t + h) = x(t) + \frac{h}{2}(F_1 + F_2)$$

Runge-Kutta Method

Alternative Formulas

$$\Rightarrow w_1 + w_2 = 1, \quad \alpha w_2 = 0.5, \quad \beta w_2 = 0.5$$

another solution

Pick α any non - zero number

$$\beta = \alpha, \quad w_1 = 1 - \frac{1}{2\alpha}, \quad w_2 = \frac{1}{2\alpha}$$

RK2 Formulas (select $\alpha \neq 0$)

$$K_1 = h f(t, x)$$

$$K_2 = h f(t + \alpha h, x + \alpha K_1)$$

$$x(t + h) = x(t) + \left(1 - \frac{1}{2\alpha}\right) F_1 + \frac{1}{2\alpha} F_2$$

Runge-Kutta Method

Fourth Order Runge-Kutta (RK4)

$$K_1 = h f(t, x)$$

$$K_2 = h f\left(t + \frac{1}{2}h, x + \frac{1}{2}K_1\right)$$

$$K_3 = h f\left(t + \frac{1}{2}h, x + \frac{1}{2}K_2\right)$$

$$K_4 = h f(t + h, x + K_3)$$

$$x(t + h) = x(t) + \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

Second order Runge-Kutta Method

Example

Solve the following system to find $x(1.02)$ using RK2

$$\dot{x}(t) = 1 + x^2(t) + t^3, \quad x(1) = -4, h = 0.01$$

Second order Runge-Kutta Method

Example

Solve the following system to find $x(1.02)$ using RK2

$$\dot{x}(t) = 1 + x^2(t) + t^3, \quad x(1) = -4, h = 0.01$$

STEP1:

$$K_1 = h f(t, x) = 0.01(1 + x^2 + t^3) = 0.18$$

$$K_2 = h f(t + h, x + K_1) = 0.01(1 + (x + 0.18)^2 + (t + .01)^3) = 0.1662$$

$$x(1 + 0.01) = x(1) + \frac{1}{2}(K_1 + K_2) = -4 + \frac{1}{2}(0.18 + 0.1662) = -3.8269$$

Second order Runge-Kutta Method

Example

STEP 2

$$K_1 = h f(t, x) = 0.01(1 + x^2 + t^3) = 0.1668$$

$$K_2 = h f(t + h, x + K_1) = 0.01(1 + (x + 0.1668)^2 + (t + .01)^3) = 0.1546$$

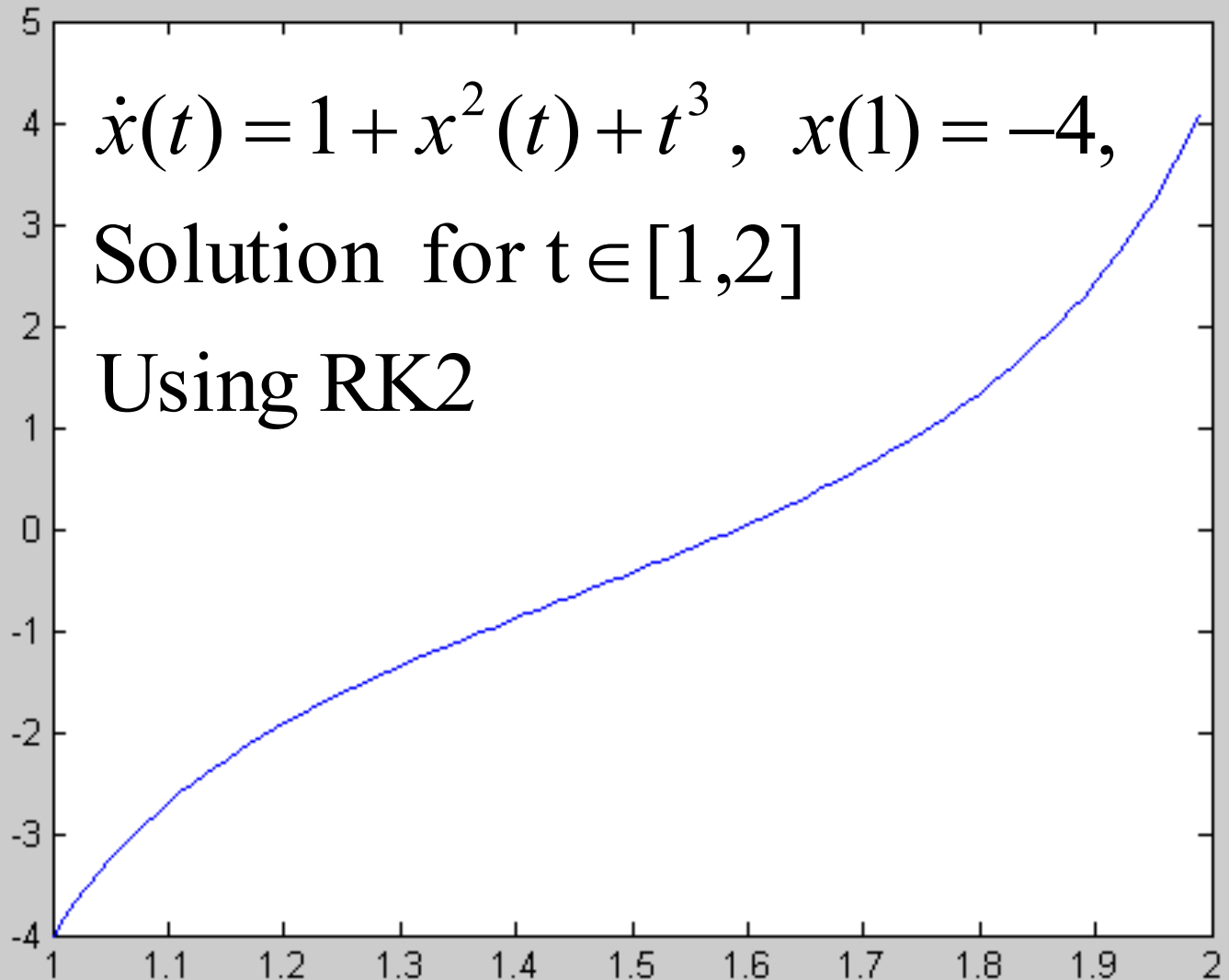
$$x(1.01 + 0.01) = x(1.01) + \frac{1}{2}(K_1 + K_2)$$

$$= -3.8269 + \frac{1}{2}(0.1668 + 0.1546) = -3.6662$$

$$\dot{x}(t) = 1 + x^2(t) + t^3, \quad x(1) = -4,$$

Solution for $t \in [1, 2]$

Using RK2



Runge-Kutta Methods

RK3

Third Order Runge Kutta (RK3)

$$K_1 = f(x_i, y_i)$$

$$K_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}K_1h\right)$$

$$K_3 = f(x_i + h, y_i - K_1h + 2K_2h)$$

$$y(x+h) = y(x) + \frac{1}{6}(K_1 + 4K_2 + K_3)$$

Runge-Kutta Methods

RK4

Fourth Order Runge Kutta (RK4)

$$K_1 = f(x_i, y_i)$$

$$K_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}K_1h\right)$$

$$K_3 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}K_2h\right)$$

$$K_4 = f(x_i + h, y_i + K_3h)$$

$$y_{i+1} = y_i + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

Fifth Order Runge-Kutta Methods

$$K_1 = f(x_i, y_i)$$

$$K_2 = f(x_i + \frac{1}{4}h, y_i + \frac{1}{4}K_1h)$$

$$K_3 = f(x_i + \frac{1}{4}h, y_i + \frac{1}{8}K_1h + \frac{1}{8}K_2h)$$

$$K_4 = f(x_i + \frac{1}{2}h, y_i - \frac{1}{2}K_2h + K_3h)$$

$$K_5 = f(x_i + \frac{3}{4}h, y_i + \frac{3}{16}K_1h + \frac{9}{16}K_4h)$$

$$K_6 = f(x_i + h, y_i - \frac{3}{7}K_1h + \frac{2}{7}K_2h + \frac{12}{7}K_3h - \frac{12}{7}K_4h + \frac{8}{7}K_5h)$$

$$y_{i+1} = y_i + \frac{h}{90}(7K_1 + 32K_3 + 12K_4 + 32K_5 + 7K_6)$$

Second Order Runge-Kutta Method

Given :

$$\frac{dy}{dx} = f(x, y)$$

$$y(x_0) = y_0$$

h

Determine # of steps needed

Second Order Runge-Kutta Method

Given :

$$\frac{dy}{dx} = f(x, y)$$

$$y(x_0) = y_0$$

h

Determine # of steps needed

RK2 formula

$$K_1 = f(x_i, y_i)$$

$$K_2 = f(x_i + h, y_i + K_1 h)$$

$$y_{i+1} = y_i + \frac{h}{2}(K_1 + K_2)$$

Second Order Runge-Kutta Method

Given :

$$\frac{dy}{dx} = f(x, y)$$

$$y(x_0) = y_0$$

h

Determine # of steps needed

RK2 formula

$$K_1 = f(x_i, y_i)$$

$$K_2 = f(x_i + h, y_i + K_1 h)$$

$$y_{i+1} = y_i + \frac{h}{2}(K_1 + K_2)$$

Step 1 :

$$K_1 = f(x_0, y_0)$$

$$K_2 = f(x_0 + h, y_0 + K_1 h)$$

$$y_1 = y_0 + \frac{h}{2}(K_1 + K_2)$$

Step 2 : $x_1 = x_0 + h$

$$K_1 = f(x_1, y_1)$$

$$K_2 = f(x_1 + h, y_1 + K_1 h)$$

$$y_2 = y_1 + \frac{h}{2}(K_1 + K_2)$$

Example 1

Second Order Runge-Kutta Method

Use the second order Runge Kutta method to solve the differential equation

$$\frac{dy}{dx} = 1 + y^2 + x^3, \quad y(1) = -4$$

to find $y(1.01), y(1.02)$

Example 1

Second Order Runge-Kutta Method

Problem:

$$\frac{dy}{dx} = 1 + y^2 + x^3, \quad y(1) = -4$$

Use RK2 to find $y(1.01), y(1.02)$



$$h = 0.01$$

$$f(x, y) = 1 + y^2 + x^3$$

$$x_0 = 1, \quad y_0 = -4$$

Example 1

Second Order Runge-Kutta Method

Problem:

$$\frac{dy}{dx} = 1 + y^2 + x^3, \quad y(1) = -4$$

Use RK2 to find $y(1.01), y(1.02)$



$$h = 0.01$$

$$f(x, y) = 1 + y^2 + x^3$$

$$x_0 = 1, \quad y_0 = -4$$

Step 1:

$$K_1 = f(x_0, y_0) = (1 + y_0^2 + x_0^3) = 18.0$$

$$K_2 = f(x_0 + h, y_0 + K_1 h) = (1 + (y_0 + 0.18)^2 + (x_0 + .01)^3) = 16.62$$

$$y_1 = y_0 + \frac{h}{2}(K_1 + K_2) = -4 + \frac{0.01}{2}(18 + 16.62) = -3.8269$$

Example 1

Second Order Runge-Kutta Method

Problem:

$$\frac{dy}{dx} = 1 + y^2 + x^3, \quad y(1) = -4$$

Use RK2 to find $y(1.01), y(1.02)$



$$h = 0.01$$

$$f(x, y) = 1 + y^2 + x^3$$

$$x_1 = 1.01, \quad y_1 = -3.8269$$

Example 1

Second Order Runge-Kutta Method

Problem:

$$\frac{dy}{dx} = 1 + y^2 + x^3, \quad y(1) = -4$$

Use RK2 to find $y(1.01), y(1.02)$



$$h = 0.01$$

$$f(x, y) = 1 + y^2 + x^3$$

$$x_1 = 1.01, \quad y_1 = -3.8269$$

Step 2:

$$K_1 = f(x_1, y_1) = (1 + y_1^2 + x_1^3) = 16.68$$

$$K_2 = f(x_1 + h, y_1 + K_1 h) = (1 + (y_1 + 0.1668)^2 + (x_1 + .01)^3) = 15.46$$

$$y_2 = y_1 + \frac{h}{2}(K_1 + K_2) = -3.8269 + \frac{0.01}{2}(16.68 + 15.46) = -3.6662$$

Example 1

Summary of the solution

Problem:

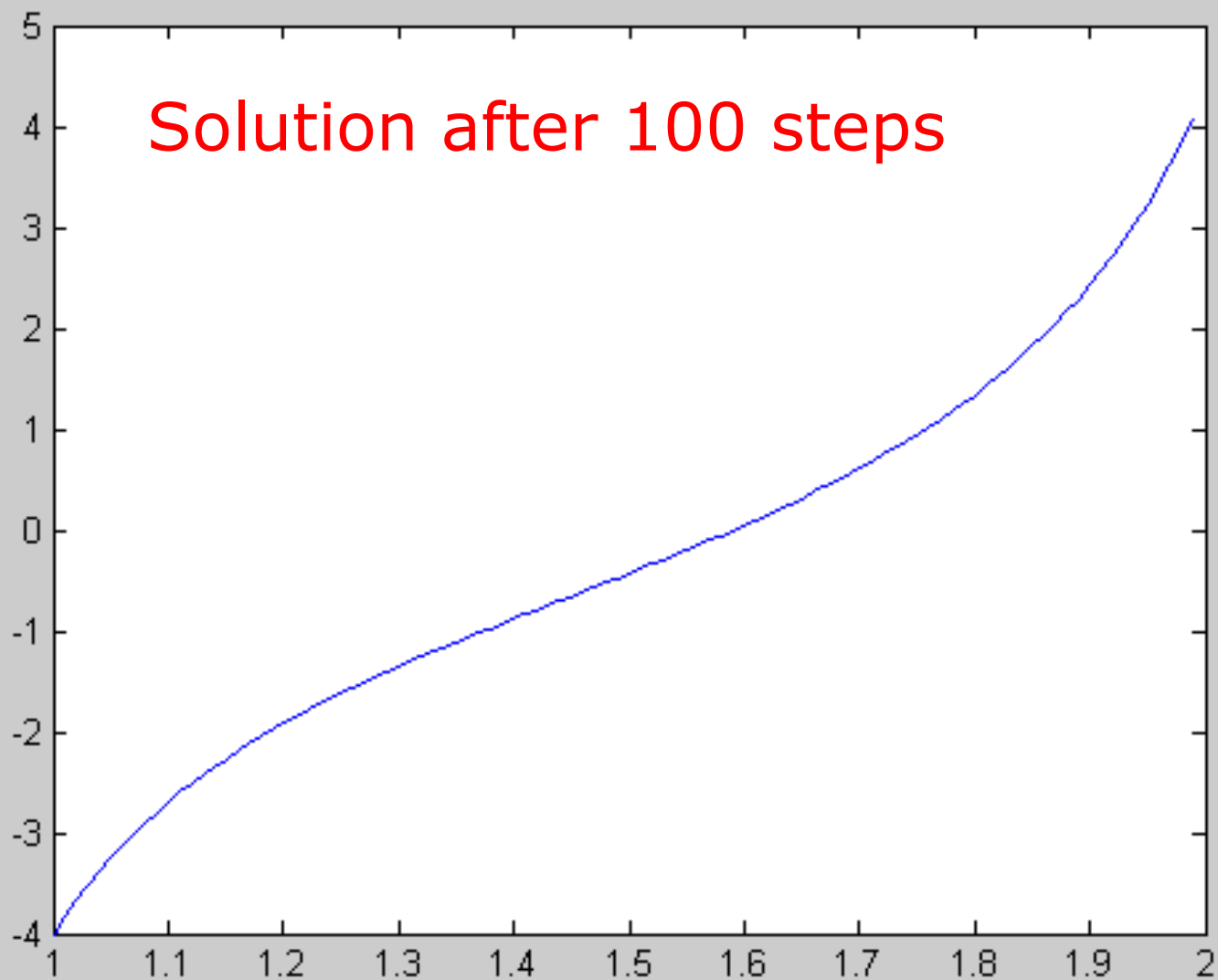
$$\frac{dy}{dx} = 1 + y^2 + x^3, \quad y(1) = -4$$

Use RK2 to find $y(1.01), y(1.02)$

Summary of the solution

i	x_i	y_i
0	1.00	-4.0000
1	1.01	-3.8269
2	1.02	-3.6662

Solution after 100 steps



Example 2

4th-Order Runge-Kutta Method

See RK4 Formula

$$\frac{dy}{dx} = 1 + y + x^2$$

$$y(0) = 0.5$$

$$h = 0.2$$

Use RK4 to compute $y(0.2)$ and $y(0.4)$

Example 2

Fourth Order Runge-Kutta Method

Problem:

$$\frac{dy}{dx} = 1 + y + x^2, \quad y(0) = 0.5$$

Use RK4 to find $y(0.2), y(0.4)$

Example 2

Fourth Order Runge-Kutta Method

See RK4 Formula

Problem:

$$\frac{dy}{dx} = 1 + y + x^2, \quad y(0) = 0.5$$

Use RK4 to find $y(0.2), y(0.4)$



$$h = 0.2$$

$$f(x, y) = 1 + y + x^2$$

$$x_0 = 0, \quad y_0 = 0.5$$

Step1:

$$K_1 = f(x_0, y_0) = 1 + y_0 + x_0^2 = 1.5$$

$$K_2 = f\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}K_1h\right) = 1 + (y_0 + 0.15) + (x_0 + 0.1)^2 = 1.64$$

$$K_3 = f\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}K_2h\right) = 1 + (y_0 + 0.164) + (x_0 + 0.1)^2 = 1.654$$

$$K_4 = f(x_0 + h, y_0 + K_3h) = 1 + (y_0 + 0.16545) + (x_0 + 0.2)^2 = 1.7908$$

$$y_1 = y_0 + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4) = 0.8293$$

Runge-Kutta Methods

RK4

Fourth Order Runge Kutta (RK4)

$$K_1 = f(x_i, y_i)$$

$$K_2 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}K_1h\right)$$

$$K_3 = f\left(x_i + \frac{1}{2}h, y_i + \frac{1}{2}K_2h\right)$$

$$K_4 = f(x_i + h, y_i + K_3h)$$

$$y_{i+1} = y_i + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

Example 2

Fourth Order Runge-Kutta Method

Problem:

$$\frac{dy}{dx} = 1 + y + x^2, \quad y(0) = 0.5$$

Use RK4 to find $y(0.2), y(0.4)$



$$h = 0.2$$

$$f(x, y) = 1 + y + x^2$$

$$x_0 = 0.2, \quad y_0 = 0.8293$$

Step 2:

$$K_1 = f(x_1, y_1) = 1.7893$$

$$K_2 = f\left(x_1 + \frac{1}{2}h, y_1 + \frac{1}{2}K_1h\right) = 1.9182$$

$$K_3 = f\left(x_1 + \frac{1}{2}h, y_1 + \frac{1}{2}K_2h\right) = 1.9311$$

$$K_4 = f(x_1 + h, y_1 + K_3h) = 2.0555$$

$$y_2 = y_1 + \frac{0.2}{6}(K_1 + 2K_2 + 2K_3 + K_4) = 1.2141$$

Example 2

Summary of the solution

Problem:

$$\frac{dy}{dx} = 1 + y + x^2, \quad y(0) = 0.5$$

Use RK4 to find $y(0.2), y(0.4)$

Summary of the solution

i	x_i	y_i
0	0.0	0.5
1	0.2	0.8293
2	0.4	1.2141

Summary

- ❑ RK methods generate an accurate solution without the need to calculate high order derivatives.
- ❑ Second order RK have local truncation error of order $O(h^3)$.
- ❑ Fourth order RK have local truncation error of order $O(h^5)$.
- ❑ N function evaluations are needed in the N^{th} order RK method.